

NAGPUR AMBEDKAR INTL, India

WMO: 428670

Lat: 21.092N

Lon: 79.047E

Elev: 315

StdP: 97.60

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	12.8	0.4	4.0	29.1	2.0	4.5	27.2	5.5	25.4	4.8	25.1	0.8	340	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.6	44.2	22.6	43.1	22.4	42.0	22.3	27.9	32.3	27.2	31.9	26.7	31.4	3.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.6	30.0	26.1	22.4	29.2	25.8	21.9	28.9	91.9	32.7	88.9	31.6	86.4	31.8	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.0	5.2	8.0	46.2	1.4	1.3	7.0	47.2	6.1	47.9	5.3	48.7	4.3	49.6		
			5.9	29.4	1.4	1.7	4.8	30.7	4.0	31.7	3.2	32.6	2.2	33.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	21.2	24.3	28.7	33.2	35.8	32.2	27.9	27.3	27.8	26.7	23.9	21.3
	DBStd	4.90	2.12	2.36	2.61	2.25	2.29	3.65	1.86	1.41	1.27	1.68	1.85	2.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	5	3	0	0	0	0	0	0	0	0	0	0	2
	CDD10.0	6402	346	401	581	697	801	667	556	536	534	519	416	349
	CDD18.3	3367	91	168	322	447	543	417	298	277	284	261	166	94
	CDH23.3	42823	873	1781	4323	7173	9241	6046	3113	2613	2792	2525	1471	872
Wind	WSAvg	2.0	1.3	1.6	1.8	2.3	3.0	3.2	2.9	2.6	1.9	1.3	1.0	1.1
	PrecAvg	1128	12	15	19	11	20	181	321	279	183	52	14	13
	PrecMax	1754	104	90	100	61	100	457	679	558	427	363	100	166
	PrecMin	606	0	0	0	0	0	22	105	78	7	0	0	0
	PrecStd	232	19	19	25	13	22	97	123	104	100	69	24	35
	DB	32.8	36.9	41.2	44.8	47.0	45.2	36.2	33.9	35.1	35.9	33.2	32.0	32.0
	MCWB	19.7	19.1	20.7	21.3	22.9	22.9	25.9	27.3	25.6	23.8	20.9	19.9	19.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.8	34.9	39.9	43.2	45.1	43.1	34.5	32.6	33.9	34.3	32.1	30.2	30.2
	2% MCWB	18.6	19.4	20.0	21.2	22.9	23.7	25.8	26.5	26.0	23.7	21.1	19.3	19.3
	5% DB	29.2	33.2	38.2	42.2	43.9	41.1	33.0	31.5	32.9	33.2	31.1	29.2	29.2
	5% MCWB	18.0	19.1	19.9	21.2	22.8	24.2	26.0	26.1	25.7	23.4	21.0	18.4	18.4
	10% DB	28.1	31.8	36.9	41.1	42.8	39.0	31.6	30.6	31.9	32.2	30.1	28.0	28.0
	10% MCWB	17.6	18.7	19.8	21.2	22.8	24.2	25.9	25.9	25.6	23.2	20.5	17.7	17.7
	WB	22.9	23.0	24.2	26.8	27.8	28.5	28.7	28.7	28.1	27.2	25.3	23.1	23.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.1	30.0	34.1	37.5	40.8	34.2	31.9	32.0	32.3	31.7	30.2	28.8	28.8
	2% WB	20.6	21.6	23.1	25.5	26.4	27.4	27.6	27.5	27.2	26.3	24.1	21.4	21.4
	2% MCDB	27.0	29.5	33.8	37.2	39.5	33.3	30.7	30.7	31.3	30.8	28.9	27.2	27.2
	5% WB	19.6	20.6	22.2	24.1	25.4	26.7	26.9	26.8	26.6	25.7	23.1	20.1	20.1
	5% MCDB	26.6	29.2	33.1	36.1	38.5	32.9	30.3	30.0	30.7	30.0	27.9	25.8	25.8
	10% WB	18.6	19.8	21.3	22.9	24.6	26.1	26.4	26.2	26.1	25.0	22.1	19.1	19.1
	10% MCDB	25.8	29.1	32.4	36.1	37.4	32.2	29.6	29.2	30.1	29.2	26.7	25.3	25.3
Mean Daily Temperature Range	MDBR	14.4	14.7	15.3	15.4	13.6	10.1	6.3	6.0	8.0	11.6	13.5	14.5	14.5
	5% DB	15.7	16.4	17.3	16.3	14.5	12.7	8.7	7.8	10.0	13.2	14.3	15.7	15.7
	5% MCWB	6.7	6.3	6.1	5.5	5.0	3.9	3.2	3.5	4.1	5.1	5.8	6.7	6.7
	5% WB	13.6	14.0	14.6	15.0	13.8	10.1	7.2	7.0	8.4	9.6	11.0	12.8	12.8
	MCWBR	7.1	6.6	5.9	6.2	5.5	3.9	3.4	3.6	4.0	4.4	5.6	6.8	6.8
	taub	0.463	0.478	0.520	0.552	0.584	0.635	0.591	0.555	0.540	0.517	0.507	0.463	0.463
	taud	2.029	2.012	1.935	1.902	1.861	1.797	1.906	2.014	2.035	2.032	1.980	2.048	2.048
Clear-Sky Solar Irradiance	Ebn at Noon	796	813	796	775	744	701	733	763	769	768	748	780	780
	Edh at Noon	159	171	192	201	208	219	197	177	171	165	166	151	151
	RadAvg	4.57	5.42	6.26	6.91	6.92	5.20	3.70	3.79	4.80	5.18	4.69	4.42	4.42
All-Sky Solar Radiation	RadStd	0.22	0.24	0.31	0.22	0.17	0.54	0.52	0.42	0.43	0.31	0.25	0.17	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.47	N/A	N/A	+0.77	+0.86	+0.76	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page